

Pathwidth and Searching in Parameterized Threshold Graphs

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Abstract. Treewidth and pathwidth are important graph parameters that represent how close the graph is to trees and paths respectively. We calculate treewidth and pathwidth on parameterized chordal and threshold graphs. We define a *chordal* + $1v$ graph as a graph that can be made into a chordal graph by removing a vertex. We give polynomial time algorithms for computing the treewidth of a *chordal* + $1v$ graph, pathwidth of a *threshold* + $1v$ graph and a *threshold* + $2e$ graph. The mixed search number of a graph is the minimum number of cops required to capture a single robber, who is hiding in the graph. We apply the algorithm to compute the pathwidth in order to compute the mixed search number of a *threshold* + $1v$ graph.

Keywords: Graph searching, treewidth, pathwidth, parameterization, threshold graphs.

1 Introduction

Treewidth is an important graph parameter which measures the tree-like nature of a graph. It is closely related to *chordal* graphs. Similarly, pathwidth measures path-like nature of a graph. Pathwidth is closely related to *interval* graphs. Computing treewidth and pathwidth of a general graph is known to be NP-complete [1]. Dynamic programming techniques on tree decompositions have allowed significant improvements on some graph algorithms like HAMILTONIAN CIRCUIT, INDEPENDENT SET, VERTEX COVER, etc. Each problem that can be formulated in Monadic Second Order Logic (MSOL) can be solved in linear time on graphs of bounded treewidth. See [3] for a survey. Computing these decompositions and widths effectively for the required class of graphs is the key subproblem in these algorithms [2]. This makes it interesting to solve treewidth and pathwidth of special classes of graphs. Some instances of an NP-complete problem are polynomially solvable and some instances are difficult. Parameterization provides a

formal way to study instances that are polynomially solvable and instances that are difficult. See [7] for a complete analysis. A *parameterization* of an alphabet Σ^* , is a mapping $k : \Sigma^* \rightarrow \mathbb{N}$ that is a polynomial time computable function. Graphs that become *chordal* by removing some k vertices are called *chordal + kv* graphs. This number k is considered as a *parameterization* of the input graph. Graph theoretic problems on graphs like *split + kv* graphs, which are also called parameterized graph classes have been studied in the past [4,13,16].

Graph searching is a scenario where cops need to capture one robber who hides in the vertices of the graph. Both the robber and the cops can move with infinite speed along the edges of the graph. The robber is invisible to the cops, but the cops are visible to the robber. However, the cops can jump between vertices in the graph and obstruct the movement of the robber. Mixed search number of a graph G , $miam(G)$ is the minimum number of cops required to capture this robber hiding in G . It has been studied extensively [12,11,9]. In a computer network setting, the graph searching problem serves as a mathematical model for protecting networks against viruses and other unwanted agents, like spyware or eavesdroppers. A practical example is the problem of finding a successful strategy for a group of collaborating software programs that are designed to clean the network from a virus [6].

In our paper, we prove in section 3 that the polynomial time algorithm for treewidth of *split + 1v* graphs [15] given by Federico Mancini, can be used for computing treewidth of *chordal + 1v* graphs. In section 4 we give a polynomial time algorithm which computes pathwidth of *threshold + 1v* graphs by modifying the interval ordering of the threshold part of the graph. Threshold graphs have many important mathematical properties and several applications like psychology, scheduling and synchronization of parallel processes [14]. Similarly we give a polynomial time algorithm for pathwidth of *threshold + 2e* graphs. Finally, we calculate $miam(G)$ for *threshold + 1v* graphs in section 6 by searching for a special kind of path decomposition called *good path decomposition*, which was introduced by Pinar Heggernes and Rodica Mihai [11].

2 Notations and Definitions

A simple graph is a collection of vertices V and edges E represented as $G = (V, E)$, where each edge is an unordered pair of distinct vertices. The set of neighbors of a vertex v is denoted by $N(v)$. The *induced subgraph* of G by a vertex set $S \subseteq V$ is $G(S) = (S, E')$ where $E' = \{(u, v) \in E \mid u \in S \text{ and } v \in S \text{ and } u \neq v\}$. A class of graphs $\mathbb{F}$ is called *hereditary* if $G = (V, E) \in \mathbb{F} \Rightarrow \forall S \subseteq V, G(S) \in \mathbb{F}$. Number $\omega(G)$ represents maximum clique size of G . A vertex v is called *simplicial* if $N(v)$ forms a clique in the graph. A graph $G = (V, E)$ is called *chordal* or *triangulated* if every cycle in G with more than 3 vertices has a chord. A minimum set of edges F to be added to a graph G to make it a chordal graph G' is called a set of *fill edges*. The graph G' is called a *minimum chordal completion* of G . A graph $G = (V, E)$ is called *split* if and only if $V = K \cup I$ such that $K \cap I = \phi$ where K is a clique in G and I is an independent set in G . A graph